



1. What is the value of $201 \times 9 + 20 \times 19 - 2 \times (0 + 1 + 9)$?

- (A) 1409 (B) 1449 (C) 2019 (D) 2169 (E) 2209

2. The value of $\frac{111,111,111}{9}$ is an integer that consists of exactly 8 distinct digits from the set $\{1, 2, \dots, 8, 9\}$. What is the unused digit?

- (A) 1 (B) 2 (C) 5 (D) 8 (E) 9

3. Let N be the unique positive integer such that the $N\%$ of 880 is a perfect square smaller than 880. What is the sum of the digits of N ?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

4. The ratio of the area of an octagon to its perimeter is $5/4$, when calculations are done in inches. When calculations are done in feet, what is the ratio of the area of the same octagon to its perimeter?

- (A) $\frac{5}{576}$ (B) $\frac{5}{48}$ (C) $\frac{5}{4}$ (D) 15 (E) 180

5. Samuel writes down the 65^{th} smallest positive integer that can be expressed as the sum of 18 consecutive integers. If the largest of those integers is n , find n .

- (A) 73 (B) 74 (C) 82 (D) 83 (E) 235

6. Ten gangsters are standing on a flat surface with the distances between them all distinct. At twelve o'clock, when the church bells start chiming, each of them fatally shoots the one among the other nine gangsters who is the farthest. At most how many gangsters will remain alive?

(A) 1 (B) 2 (C) 7 (D) 8 (E) 9

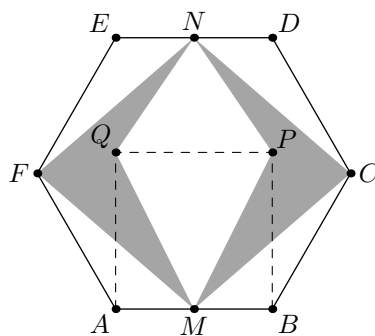
7. We define a real number x to be a *semi-integer* if $\sqrt{2} \cdot x$ is an integer. How many real numbers $0 \leq x \leq 100$ are *semi-integers*?

(A) 71 (B) 140 (C) 141 (D) 142 (E) 211

8. For real numbers m and n , we define the operator $\circ$ as $m \circ n = m^{-1} - (mn)^{-1} + n^{-1}$. What is the value of $1 \circ (2 \circ (\cdots (2018 \circ 2019)))$?

(A) $\frac{1}{2019}$ (B) $\frac{2018}{2019}$ (C) 1 (D) $\frac{2019}{2018}$ (E) 2019

9. Regular hexagon $ABCDEF$ has side length 1. Points P and Q are placed in the interior of the hexagon such that $ABPQ$ is a square. If M and N are the midpoints of sides AB and DE , respectively, what is the area of the shaded region?



(A) $\frac{2}{3}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{8}{9}$ (D) 1 (E) $\frac{2\sqrt{3}}{3}$

10. Consider a polynomial $p(x)$ of degree 1 such that for a real number a , $p(a) = 2$, $p(p(a)) = 17$ and $p(p(p(a))) = 167$. What is the value of a ?

(A) -1 (B) $-\frac{1}{2}$ (C) 0 (D) $\frac{1}{2}$ (E) 1

11. Let $0 < x < 100$ be a randomly chosen real number. What is the expected value of $\lfloor x \rfloor \{x\}$?

(A) 24.25 (B) 24.75 (C) 25 (D) 49.5 (E) 50

12. Geoff the frog is standing at the origin in the coordinate plane. For each move, Geoff can only move one unit to the right or one unit upwards; also, every up move must be immediately followed by a right move (except for the last move). What is the number of distinct sequences of moves that end at the point $(9, 5)$?

- (A) 56 (B) 126 (C) 210 (D) 252 (E) 346

13. Let $\mathcal{C}$ be a circle with radius 10, and let $1 \leq r \leq 9$ be a randomly chosen positive integer. Derek randomly picks a point P lying inside $\mathcal{C}$ and constructs another circle $\mathcal{C}'$ centered at P with radius r . If the probability that the two circles intersect can be written as $\frac{m}{n}$ for relatively prime positive integers m and n , what is the value of $m + n$?

- (A) 61 (B) 72 (C) 79 (D) 83 (E) 101

14. If $a, b > 1$ satisfy $\log_4(a) + \log_9(a) = \log_a(4) + \log_a(9)$ and $\log_9(b) + \log_{16}(b) = \log_b(9) + \log_b(16)$, what is the value of $\log_a(b)$?

- (A) $\frac{1}{2}$ (B) $\frac{\sqrt{2}}{2}$ (C) 1 (D) $\sqrt{2}$ (E) 2

15. In triangle $\triangle ABC$, let D be the midpoint of BC and E be a point on AD so that $\angle BEC + \angle BAC = 180^\circ$. If $BC = \sqrt{7}$, the value of $DE \cdot DA$ can be written as $\frac{m}{n}$ for some relatively prime integers m, n . What is $m + n$?

- (A) 7 (B) 9 (C) 11 (D) 51 (E) 53

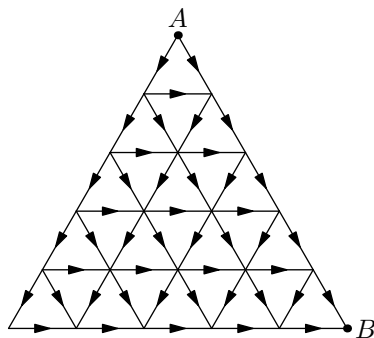
16. A random number generator generates a random number each second. On the n^{th} second, the generator randomly outputs an integer from the set $\{1, 2, \dots, n\}$. What is the number of seconds it will take before the expected sum of the outputted numbers is at least 280?

- (A) 23 (B) 24 (C) 30 (D) 32 (E) 35

17. Let $\triangle ABC$ be an equilateral triangle with its three vertices on the graph $y = |2x|$. If the midpoint of AC is $(3, 6)$, then what is the area of $\triangle ABC$?

- (A) $20\sqrt{3}$ (B) $25\sqrt{3}$ (C) $\frac{80\sqrt{3}}{3}$ (D) $\frac{85\sqrt{3}}{3}$ (E) $45\sqrt{3}$

18. James must travel from point A to point B along the grid shown below. He can only move along the arrows and he must take exactly 8 steps. How many possible paths are there?



- (A) 70 (B) 72 (C) 140 (D) 360 (E) 560

19. How many ordered triples of primes (p, q, r) satisfy $p^2 + q^2 = pqr + 1$?

- (A) 2 (B) 4 (C) 6 (D) 7 (E) 8

20. Let $\mathcal{T}'$ be the projection on a horizontal plane of an equilateral triangle $\mathcal{T}$ with side 5. If two of the sides of $\mathcal{T}'$ have length 3 and 4, what is the length of the third side?

- (A) 3 (B) $2\sqrt{3}$ (C) $3\sqrt{2}$ (D) $2\sqrt{6}$ (E) 5

21. For positive integers a and b , let $\star$ be an operator with the following properties: $a \star b = b \star a$; $(a + 1) \star b = a \star b + a + b$; $1 \star 1 = 2$. Let c be the unique positive integer such that there exists exactly 25 ordered pairs of positive integers (a, b) satisfying $a \star b = c$. What is the value of c ?

- (A) 301 (B) 326 (C) 351 (D) 376 (E) 401

22. Let $\Gamma_1, \Gamma_2, \Gamma_3$ be three circles with radii 2, 3, 6 respectively, such that Γ_1 and Γ_2 are externally tangent at A and Γ_3 is internally tangent to Γ_1 and Γ_2 at B, C respectively. What is the radius of the circumcircle of $\triangle ABC$?

- (A) $\frac{6}{\sqrt{11}}$ (B) 5 (C) $\sqrt{2} + \sqrt{3} + \sqrt{6}$ (D) 6 (E) $4\sqrt{3}$

23. Suppose

$$\frac{1}{\tan 1^\circ + \cot 1^\circ} + \frac{1}{\tan 2^\circ + \cot 2^\circ} + \cdots + \frac{1}{\tan 44^\circ + \cot 44^\circ} = \frac{\cot 1^\circ - a}{b}$$

for positive integers a and b . What is the value of $|a - b|$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

24. Blaise creates a circle of 100 unlit candles and randomly lights ten of them. Afterwards, every second, each unlit candle has a $k/2$ chance of becoming lit, where k is the number of lit candles adjacent to the unlit candle. Which of these is nearest to the average number of seconds that pass before all 100 candles are lit?

- (A) 90 (B) 95 (C) 100 (D) 150 (E) 200

25. Let $\triangle ABC$ be a triangle with incenter I , and let P be the intersection of line AI with the circumcircle of $\triangle ABC$. Let points X and Y denote the incenters of $\triangle ABP$ and $\triangle APC$, respectively. Suppose that $XY = 2$ and $BI^2 + CI^2 = 15$. The value of $\cos \angle BAC$ can be written as $\frac{p}{q}$ for some relatively prime integers p and q . What is the value of $p + q$?

- (A) 7 (B) 49 (C) 57 (D) 79 (E) 81

Time limit: 75 minutes